3 Points Each

2000

1. There are 29 students in a class, and the number of girls is 3 greater than the number of boys. How many girls are in this class?

(A) 6
(B) 13
(C) 16
(D) 19
(E) 15

2. The number -11 – 2 (-7) is equal to

(A) 3
(B) -3
(C) -25
(D) 25
(E) 16

3. There are 585 drawers in an old, huge chest. In each drawer there are 3 mice, and each one is the mother of 5 little mouse pups, which also live with them. How many little mouse pups are there in the chest?

(A) (585 + 3) +3
(B) (585 x 3) + 5
(C) (585 x 5) + 3
(D) 585 x 3 x 5
(E) 585 x (3+5)

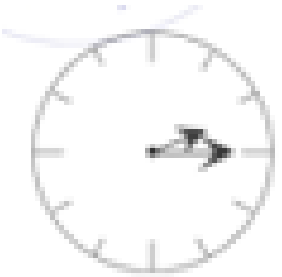
4. The sum of five consecutive natural numbers is equal to 2000. The greatest of these numbers is

(A) 490
(B) 475
(C) 471
(D) 423
(E) 402

5. A train is 56 km away from the nearest station and is approaching the station traveling 9 km for every 10 minutes. How far from the station will the train be after 30 minutes?

(A) 47 km
(B) 39 km
(C) 31 km
(D) 29 km
(E) 26 km

6. The picture shows the mirror reflection of a wall clock. What time does the actual clock show?



(A) 3:15
(B) 10:15
(C) 10:45
(D) 2:15
(E) 9:45

7. It is now the year 2000. How many twos and fives are there among the prime factors of the number 2000?

(A) 2 twos and 5 fives
(B) 3 twos and 3 fives
(C) 3 twos and 4 fives
(D) 4 twos and 3 fives
(E) 4 twos and 4 fives

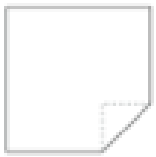
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8. Anna's birthday present is placed in a box with dimensions 10 cm x 10 cm x 30 cm and wrapped with a ribbon as shown in the picture. What is the length of the ribbon (ignore the length needed to make knots)?



(A) 200 cm
(B) 240 cm
(C) 260 cm
(D) 300 cm
(E) 250 cm

9. One of the corners of a square was folded (see the picture). Two other corners of the square were then folded the same way. How many vertices does the figure formed in this way have?



(A) 3
(B) 4
(C) 5
(D) 6
(E) 7

10. The kangaroo is facing towards X (see the picture). Which letter will the kangaroo face if it rotates 270° clockwise?



(A) A
(B) B
(C) C
(D) D
(E) E

4 Points Each

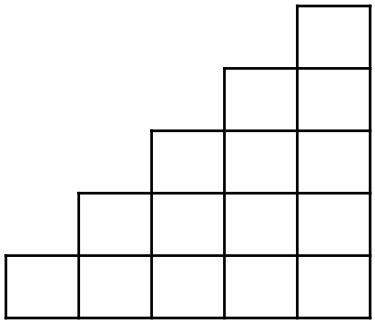
11. How many two-digit numbers are divisible by both 2 and 7?

(A) 8
(B) 7
(C) 6
(D) 5
(E) 4

12. If A – 1 = B + 2 = C – 3 = D + 4 = E – 5 , then which of the numbers A, B, C, D, and E, is the greatest?

(A) A
(B) B
(C) C
(D) D
(E) E

13. Out of how many small squares will a figure be made, if it is made the same way as the figure in the picture but has 10 steps?



(A) 25
(B) 30
(C) 40
(D) 55
(E) 100

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14. How long will it take to print one million forms, if it takes 1 minute to print 100 of these forms?

(A) 160 hours and 40 minutes
(B) 166 hours and 40 minutes
(C) 120 hours and 40 minutes
(D) 18 hours and 10 minutes
(E) 120 hours

15. Each of the five neighbors owns a rectangular plot of land with the same area. The flower gardens of their land are fenced in (solid line in the pictures). Who has the longest fence?



(A) Mr. Adam
(B) Mr. John
(C) Mr. Jack
(D) Mr. Peter
(E) Mr. Mark

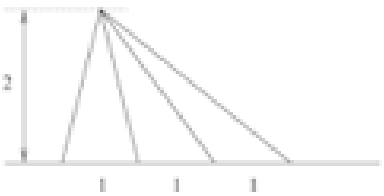
16. Number a is greater than number b. The difference between numbers a and b is 15. If we decrease number a by 5 and increase number b by 2, then the difference will

(A) decrease by 8
(B) decrease by 5
(C) increase by 8
(D) increase by 5
(E) decrease by 7

17. John comes to the computer lab every day, Karl every 2 days, Stan every 3 days, Adam every 4 days, Paul every 5 days, and Peter every 6 days. Today they are all at the computer lab. In how many days will they all be there together again?

(A) 6 days
(B) 20 days
(C) 30 days
(D) 60 days
(E) 90 days

18. What is the sum of the areas of all the triangles you can see in the picture to the right?



(A) 4
(B) 5
(C) 7
(D) 8
(E) 10

19. Which four digits need to be removed from the number 4921508 to get the smallest possible three-digit number?

(A) 4, 9, 2, 1
(B) 4, 2, 1, 0
(C) 1,5,0,8
(D) 4, 9, 2, 5
(E) 4,9,5,8

20. How many four-digit numbers with a sum of their digits equal to 3 are there?

(A) 6
(B) 8
(C) 9
(D) 10
(E) 12

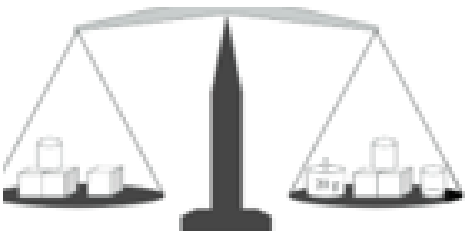
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5 Points Each

21. The leaders of the math camp in Zakopane decided to divide 96 participants into equal groups where the number of participants in each group is between 5 and 20. In how many ways can this be done?

(A) 10
(B) 8
(C) 5
(D) 4
(E) 2

22. The scale shown in the picture is in balance. On the scales there is a 20 g weight and certain solids: cubes and cylinders. All the solids (cubes and cylinders) together weigh 500 g. How much does one cube weigh?

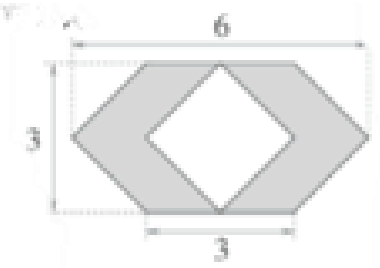


(A) 40 g
(B) 50 g
(C) 60 g
(D) 70 g
(E) 80 g

23. The length of one of the sides of a rectangle was increased by 10%, and the length of the other side of the rectangle was decreased by 10%. How did the area of the rectangle change?

(A) It did not change.
(B) It decreased by 1%.
(C) It increased by 1%.
(D) It increased by 20%.
(E) It depends on the lengths of the sides.

24. What is the area of the shaded figure (see the picture to the right)?



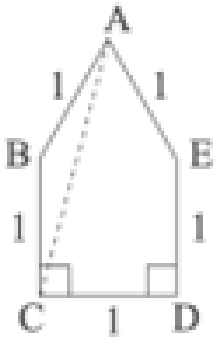
(A) 5
(B) 9
(C) 12
(D) 15
(E) 18

25. The jump of a little kangaroo is 1 meter long and takes one-half of a second. His mother's jump is 3 meters long and takes one second. The mother and the little kangaroo start at the same time from the same place and are jumping towards a eucalyptus which is 180 meters away. For how many seconds will the mother be waiting for the little kangaroo at the eucalyptus?

(A) 30
(B) 60
(C) 10
(D) 120
(E) 20

26. In the pentagon ABCDE (see the picture), the measure of angle BAC is

(A) 15°
(B) 12°
(C) 30°
(D) 20°
(E) other



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27. How many different counterweights can be found to balance a scale using any or all of the following three weights: 1 kg, 3 kg, and 9 kg? (We can place any weight on any side.)

(A) 3
(B) 6
(C) 11
(D) 13
(E) 14

28. We used 8 grams of playdough to make a cube with an edge of 2 cm. How many grams of playdough do we need to make a cube with an edge of 4 cm?

(A) 16
(B) 24
(C) 32
(D) 48
(E) 64

29. The body of a certain caterpillar is made up of five spherical parts, 3 of which are yellow and 2 are green. What is the greatest possible number of different types of this caterpillar that could exist?



(A) 6
(B) 8
(C) 9
(D) 10
(E) 12

30. We have 3 boxes in one row: one red, one green, and one blue. We also have 3 objects: a coin, a shell, and a bead. In each of the boxes there is only one of the objects. We know that:

- the green box is to the left of the blue box;
- the coin is to the left of the bead;
- the red box is to the right of the shell; and
the bead is to the right of the red box.

In which box is the coin?

(A) in the red box
(B) in the green box
(C) in the blue box
(D) This cannot be determined.
(E) The conditions given above cannot all be true at the same time.

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Answers key:

1. C
2. A
3. D
4. E
5. D
6. E
7. D
8. B
9. E
10. E
11. B
12. E
13. D
14. B
15. C
16. E
17. D
18. E
19. D
20. D
21. D
22. D
23. B
24. B
25. A
26. A
27. D
28. E
29. D
30. A

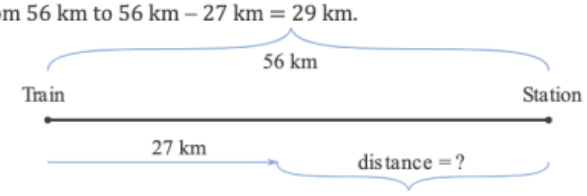
1. (C) 16
To the class of 29 students add 3 new boys to equal the number of boys and girls. Then this new class has 32 students and half of them are girls. There are $(29 + 3) \div 2 = 16$ girls in the class.

2. (A) 3
$-11 - 2(-7) = -11 - (-14) = -11 + 14 = 3$

3. (D) $585 \times 3 \times 5$
There are 3 mice in each drawer, so there are (585×3) mice. Each mouse has 5 little mouse pups, so there are $(585 \times 3) \times 5 = 585 \times 3 \times 5$ little mouse pups in the chest.

4. (E) 402
Let m be the middle number of the five consecutive numbers. Then the five numbers are $m-2$, $m-1$, m, $m+1$, $m+2$. Their sum is $5m$, so $5m = 2000$ and $m = 400$.
The greatest of the five numbers is $m + 2 = 400 + 2 = 402$.

5. (D) 29 km
30 minutes $= 3 \times 10$ minutes and the train moves 9 km in 10 minutes, so in 30 minutes the train moves 3×9 km $= 27$ km. It moves towards the station, so after 30 minutes the distance will decrease from 56 km to 56 km – 27 km = 29 km.



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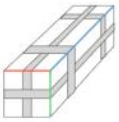
6. (E) 9:45
If the images of a wall clock and its mirror reflection are put on one clock, then the images are symmetric with respect to the vertical line passing through the center of the clock.



The hour and the minute hands of the mirror reflection point to the blue dots. The symmetry with respect to the vertical axis is applied and the results are shown in orange.
Thus, the minute hand of the actual wall clock points towards 9 indicating 45 minutes past the hour, and the hour hand points somewhere between 9 and 10 (closer to 10). Hence, the wall clock shows 9:45.

7. (D) 4 twos and 3 fives
$2000 = 2 \times 1000 = 2 \times 10^3 = 2 \times (2 \times 5)^3 = 2 \times 2^3 \times 5^3 = 2^4 \times 5^3$, so among the prime factors of 2000 there are 4 twos and 3 fives.

8. (B) 240 cm
Three faces of the box are visible. Each of them has a number of pieces of the ribbon. Add the lengths of all visible pieces and then double the sum (to account for the nonvisible faces) to get the length of the whole ribbon.

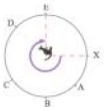


There are 3 types of visible pieces: 3 pieces that are parallel to the red edge, 3 pieces that are parallel to the green edge, and 2 pieces that are parallel to the blue edge. The dimensions of the box are 10 cm × 10 cm × 30 cm, so the 6 short pieces have lengths of 10 cm each and the 2 long pieces have a length of 30 cm each. $6 \times 10 + 2 \times 30 = 120$, so the combined length of the visible pieces is 120 cm.
The length of the whole ribbon is 2×120 cm = 240 cm.

9. (E) 7
One folded corner makes 2 vertices. Two folded corners make 4 vertices and three folded corners make 6 vertices. Thus, the final figure with 6 vertices by the three folded corners and one unfolded corner has 7 vertices.



10. (E) E
A rotation around a whole circle is 360°, so 270° is three-quarters of the way around. Going clockwise, the letter B is 90° or one-quarter turn from X, and then E is 180° or half-way around from B. After three-quarters of a turn, the kangaroo faces E.



11. (B) 7
Multiples of both 2 and 7 are exactly the multiples of 14. The two-digit multiples of 14 are 1×14, 2×14, 3×14, 4×14, 5×14, 6×14, and 7×14. $8 \times 14 = 112$, so it is not a two-digit number. There are 7 such numbers.

12. (E) E
Eliminate all subtractions from $A - 1 = B + 2 = C - 3 = D + 4 = E - 5$ by adding 5 to each expression. Then $A + 4 = B + 7 = C + 2 = D + 9 = E$, so E is greater than any other number. Hence, E is the greatest number among the five numbers.

13. (D) 55
The figure in the picture has 5 steps and also 5 small squares in the bottom row. For this figure the number of small squares is $1 + 2 + 3 + 4 + 5$.
For a figure with 10 steps the number of small squares is:
$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 =$
$= (1 + 10) + (2 + 9) + (3 + 8) + (4 + 7) + (5 + 6) = 5 \times 11 = 55$.

14. (B) 166 hours and 40 minutes
1,000,000 = 10,000 × 100, so it will take 10,000 minutes to print one million forms.
10,000 minutes = 166 × 60 minutes + 40 minutes = 166 hours and 40 minutes.

15. (C) Mr. Jack
Plots (B), (D), and (E) have the same fence length as plot (A). For each of the four plots except plot (A), we can match segments of the solid line with the segments of the dashed line as indicated below by letters u, d, *l*, r. At a given segment the letters indicate: move it up, down, left, or right, respectively.



Only for figure (C), Mr. Jack's plot, there are solid line segments (the four short horizontal segments marked in green) without dashed matches, so the length of Mr. Jack's fence is greater (by the four short horizontal segments) than any other fence.

16. (E) decrease by 7
Decrease number a by 5 to get $a - 5$. Increase number b by 2 to get $b + 2$. Compute the difference between the new numbers: $(a - 5) - (b + 2) = a - 5 - b - 2 = (a - b) - 7$.
Thus, the difference $a - b$ is decreased by 7. A numerical value of $a - b$ is not needed for this solution.

17. (D) 60 days
We need the least common multiple of 1, 2, 3, 4, 5, and 6. 30 is a multiple of 1, 2, 3, 5, and 6 but it is not a multiple of 4. We need 60 days to see John, Karl, Stan, Adam, Paul, and Peter together again at the lab.

18. (E) 10
There are three triangles with the base 1, two triangles with the base 2, and one triangle with the base 3. 2 is the common height for all these triangles, so the sum of the areas is
$\frac{1}{2} \times 2 \times (1 + 1 + 1 + 2 + 2 + 3) = 10$.

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19. (D) 4, 9, 2, 5
The smallest 3-digit numbers start with 1. This eliminates digits 4, 9, and 2 from 4921508, so we are left with 1508. Next we have to eliminate 5, 0, or 8 to get a 3-digit number starting with 1. The smallest among them is 108, so we need to remove the digit 5. Thus, 4, 9, 2, and 5 must be removed.

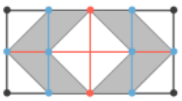
20. (D) 10
The first digit can be 1, 2, or 3. If it is 3, then the only acceptable 4-digit number is 3000.
If the first digit is 2, then the acceptable numbers are 2100, 2010, and 2001.
If the first digit is 1, then the acceptable numbers are 1200, 1020, 1002 and the numbers 1110, 1101, and 1011.
Together, there are $1 + 3 + 6 = 10$ four–digit numbers with a sum of digits equal to 3.

21. (D) 4
If 96 participants are divided into equal groups, then the number of participants in each group is a divisor of 96. We list all divisors of 96 in pairs: 1, 96; 2, 48; 3, 32; 4, 24; 6, 16; and 8, 12. Among the divisors only 6, 8, 12, and 16 are between 5 and 20. There are 4 of them, so the division into equal groups can be done in 4 ways.

22. (D) 70 g
Keeping the scale still in balance we can remove two cubes and one cylinder from each side of the scale. Thus, one cube is balanced by 20 g + one cylinder. Five cubes are balanced by 100 g + five cylinders. Together on the scale there are 5 cubes and 3 cylinders. They weigh 500 g but the weight of five cubes is 100 g + the weight of five cylinders, so 500 g = 100 g + the weight of five cylinders + the weight of three cylinders. Therefore, 400 g = the weight of eight cylinders. One cylinder weighs 50 g and one cube weighs 20 g + one cylinder, so one cube weighs 20 g + 50 g, which is 70 g.

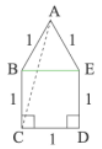
23. (B) It decreased by 1%.
Let h and w be the sides of a given rectangle. If we increase h by 10% and decrease w by 10%, then the area of the new rectangle is $(1.10 \times h) \times (.90 \times w) = .99 \times (h \times w)$, which is .01, or 1%, less than $h \times w$. Since $h \times w$ is the area of the given rectangle, the area decreased by 1%.

24. (B) 9
A rectangle (shown in black) has the width twice the size of the height. We divide each height into two equal intervals and each width into four equal intervals. All these 12 intervals have the same length, which is half of the height. The length of two such intervals is exactly the length of the height. The red lines are the axes of symmetry for the black rectangle connecting the midpoints of the opposite sides. An endpoint of the vertical red line is the midpoint of the segment with blue endpoints, so the length of this segment is equal to the length of the height. The whole rectangle is divided into 16 equal right isosceles triangles. The shaded area consists of 8 of them, so the shaded area is half of the area of the big rectangle. The original dimensions of the big rectangle are 3×6, so the shaded area is $(3 \times 6) \div 2 = 9$.



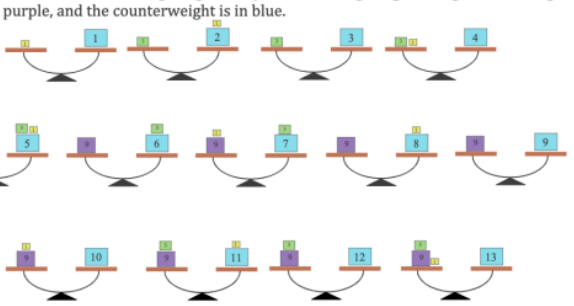
25. (A) 30
The mother needs sixty 3-meter jumps to cover 180 meters. One of her jumps takes 1 second, so after 60 seconds she reaches the eucalyptus. During these 60 seconds the little kangaroo is making 120 jumps since it needs only one-half of a second for 1 jump. Its one jump is one meter long, so after the first 60 seconds the little kangaroo is 120 meters away from the start and at the distance of (180 – 120) meters = 60 meters from the eucalyptus. The mother is waiting while the little kangaroo is covering the last 60 meters. It needed 60 seconds to cover 120 meters, so it needs 30 seconds to cover the last 60 meters. These 30 seconds is how long the mother will wait at the eucalyptus for the little kangaroo.

26. (A) 15°



BE = 1, so △BEA is an equilateral triangle and ∠ABE = 60°.
∠EBC = 90°, so ∠ABC = ∠ABE + ∠EBC = 60° + 90° = 150°.
△CAB is an isosceles triangle, so ∠BAC = (180° − 150°) ÷ 2 = 15°.

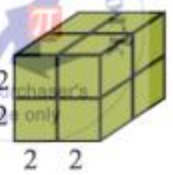
27. (D) 13
Below the counterweight numbers are in **bold**.
By using only one of the three weights we can get $1 = \mathbf{1}$, $3 = \mathbf{3}$, and $9 = \mathbf{9}$.
Placing two weights on one side and none on the other side gives us $1 + 3 = \mathbf{4}$, $1 + 9 = \mathbf{10}$, and $3 + 9 = \mathbf{12}$.
Using two weights but not placing both on one side gives us $3 - 1 = \mathbf{2}$, $9 - 1 = \mathbf{8}$, and $9 - 3 = \mathbf{6}$.
Placing all three weights on one side gives us $1 + 3 + 9 = \mathbf{13}$.
Using all three weights, but not placing all of them on one side, gives us $9 - (1 + 3) = \mathbf{5}$, $(1 + 9) - 3 = \mathbf{7}$, and $(3 + 9) - 1 = \mathbf{11}$. The numbers 1, 3, 9, 4, 10, 12, 2, 8, 6, 13, 5, 7, and 11 are different, so 13 counterweights can be found.
Below is a visual solution. The 1 kg weight is in yellow, the 3 kg weight is in green, the 9 kg weight is in purple, and the counterweight is in blue.



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28. (E) 64

A cube of the size 4 cm × 4 cm × 4 cm consists of 8 smaller cubes of the size 2 cm × 2 cm × 2 cm (see the picture). Since one smaller cube uses 8 grams of play dough, 8 × 8 grams = 64 grams of play dough is needed to make the 4 cm × 4 cm × 4 cm cube.



29. (D) 10

Let G stand for "green" and Y for "yellow." Starting with the front of caterpillar, we can color the five spherical parts by choosing where the 2 green parts are placed (the other 3 parts are yellow).

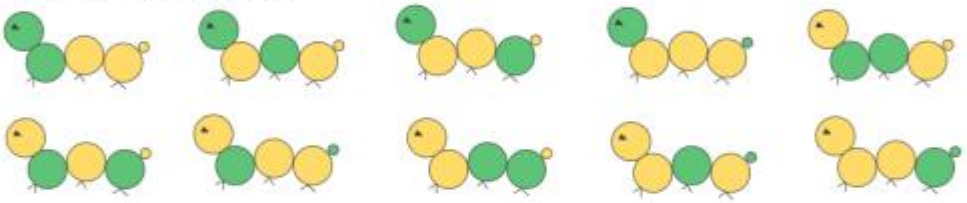


We can list the possibilities in alphabetical order as follows.

GGYYY; GYGYY, GYYGY, and GYYYG; YGGYY, YGYGY, and YGYYG; YYGGY, YYGYG; and YYYGG.

Together, there are 10 options for this type of caterpillar.

Below is a visual solution.



30. (A) in the red box

The red box is to the right of the shell, so the first box is not red.

The bead is to the right of the red box, so the last box is not red. Hence, the middle box is red.

The green box is to the left of the blue box but neither is in the middle, so the green box must be the first one and the blue box must be the last one.

Again, *the red box is to the right of the shell* and the red box is in the middle, so the shell is in the green box.

Similarly, *the bead is to the right of the red box* and the red box is in the middle, so the bead is in the blue box.

Therefore, the coin must be in the red box.

At this point, the colors and the items in the boxes are completely determined.

So far our reasoning is not making use of the information that *the coin is to the left of the bead*.

Fortunately, this is the case, so (E) is not a valid option and the solution is shown below.



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